

# Cardiac pathophysiology: real-time PCR detection of cryptosporidium parvum infection in camels

Tasa de infección de cryptosporidium parvum diagnosticada mediante técnica de PCR en tiempo real en camellos

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## Abstract

**C**ryptosporidium's significance in the realm of public health and economics lies in its capacity to induce cryptosporidiosis and resultant financial losses stemming from animal diseases. The primary objective of this investigation was to ascertain the prevalence of Cryptosporidium infection in camels inhabiting the arid regions of western Iraq, and to understand its cardiac physiology that enables it to thrive in such extremely dry conditions. This was accomplished through the isolation and diagnosis of Cryptosporidium parvum utilizing the real-time polymerase chain reaction (PCR) technique. The study, conducted in the Qadisiyah province, spanned a duration of 6 months, from November 2022 to April 2022. A total of 50 fecal samples were collected from camels, evenly distributed among both sexes, and categorized into three age groups: less than one year, 1-2 years, and 2-10 years. DNA extraction was performed using a stool DNA extraction Kit, and the obtained DNA samples underwent Real-Time PCR analysis, relying on specific primers designed for the heat shock protein gene in Cryptosporidium parvum. These primers were developed based on data from the NCBI-Genbank database (Genbank code: GQ259151.1) and primer3 plus. The findings revealed a Cryptosporidium parvum infection rate of 24%, with 12 out of 50 samples testing positive, and a remarkable Real-Time PCR efficiency of 99%. This infection was observed in both male (16.66%) and female (30.76%) camels, with a higher infection rate detected in adult camels (47.05%), followed by camels aged less than one year (14.28%), and a 10.52% infection rate in the young stage. Significant differences ( $p < 0.05$ ) were observed in relation to age and gender.

**Keywords:** Cryptosporidium, Cardiac physiology, Real-time PCR, Camel infection, Public health impact

## Resumen

**L**a importancia del Cryptosporidium en el ámbito de la salud pública y la economía radica en su capacidad para inducir criptosporidiosis y las pérdidas financieras resultantes derivadas de enfermedades animales. El objetivo principal de esta investigación fue determinar la prevalencia de la infección por Cryptosporidium en camellos que habitan las regiones áridas del oeste de Irak y comprender su fisiología cardíaca que le permite prosperar en condiciones tan extremadamente secas. Esto se logró mediante el aislamiento y diagnóstico de Cryptosporidium parvum utilizando la técnica de reacción en cadena de la polimerasa (PCR) en tiempo real. El estudio, realizado en la provincia de Qadisiyah, duró seis meses, desde noviembre de 2022 hasta abril de 2022. Se recogieron un total de 50 muestras fecales de camellos, distribuidas uniformemente entre ambos sexos, y se clasificaron en tres grupos de edad: menos de un año, 1-2 años y 2-10 años. La extracción de ADN se realizó utilizando un kit de extracción de ADN en heces y las muestras de ADN obtenidas se sometieron a análisis de PCR en tiempo real, basándose en cebadores específicos diseñados para el gen de la proteína de choque térmico en Cryptosporidium parvum. Estos cebadores se desarrollaron a partir de datos de la base de datos NCBI-Genbank (código Genbank: GQ259151.1) y primer3 plus. Los hallazgos revelaron una tasa de infección por Cryptosporidium parvum del 24 %, con 12 de 50 muestras que dieron positivo y una notable eficiencia de PCR en tiempo real del 99 %. Esta infección se observó tanto en camellos machos (16,66%) como hembras (30,76%), detectándose una mayor tasa de infección en los camellos adultos (47,05%), seguidos de los camellos menores de un año (14,28%) y un 10,52%. Tasa de infección en la etapa joven. Se observaron diferencias significativas ( $p < 0,05$ ) en relación con la edad y el sexo.

**Palabras clave:** Cryptosporidium, Fisiología cardíaca, PCR en tiempo real, Infección en camellos, Impacto en la salud pública

**E**rnest Edward Tyzzer discovered the genus *Cryptosporidium*, a coccidian parasite classified within the Apicomplexa phylum, for the first time in 1907. However, human infection was not recorded until 1976<sup>1</sup>. *Cryptosporidium parvum* is considered the causative agent of gastrointestinal and respiratory illnesses (cryptosporidiosis) in humans, domestic, and wild animals worldwide<sup>2-5</sup>. The primary symptoms include mild to severe watery diarrhea containing mucus, anorexia, lethargy, and weight loss. Cryptosporidial enteritis is usually self-limiting in healthy mammals but can be life-threatening when dehydration and other consequences of diarrhea occur in weakened immune systems<sup>6</sup>.

In addition to its public health importance, cryptosporidiosis also holds economic significance for farmers. This has been documented due to the high rate of morbidity and sometimes mortality among farm animals. Additionally, decreased performance, loss of production, and the costs resulting from interventions necessitated by diarrheic problems<sup>7</sup>.

The direct life cycle of *Cryptosporidium* begins with sporulated oocysts containing four sporozoites (the infective stage) shed in the feces of an infected host. Transmission occurs through contaminated food and water (fecal-oral route)<sup>6,8</sup>. Sporozoites are released and invade epithelial cells, where asexual schizogony reproduction occurs, followed by a sexual stage resulting in zygotes. These zygotes develop into thick-walled oocysts that are excreted from the body or thin-walled oocysts responsible for autoinfection.

Young animals of all mammals can be infected by cryptosporidiosis, such as lambs and calves. Rabbits, fish, and birds can also be infected, although it is less common in horses, cats, and dogs. Severe diarrhea is the clinical sign of cryptosporidiosis, usually appearing after 3-5 days, accompanied by anorexia, stiffness, dehydration, hyperpnoea, cessation of milk consumption, depression, slow gait, and growth retardation<sup>7,9-11</sup>.

Diagnostic methods depend on the presence of organisms in stained fecal samples after flotation or sedimentation. In cases with respiratory signs, sputum, bronchoalveolar lavage samples, and other respiratory samples may be collected. Other samples, such as intestinal

scrapings, biopsy specimens, or bile, may occasionally be necessary. Cryptosporidial antigens can be detected with Enzyme Immunosorbent Assays (ELISAs), immunochromatographic dipstick tests, and other rapid immunoassays<sup>12</sup>. Polymerase chain reaction (PCR) tests may be available in some diagnostic laboratories, either specifically for *Cryptosporidium* or as part of commercial multiplex assays for enteric pathogens<sup>13</sup>. The distribution of the disease in camels has not been properly investigated and is restricted to at least one case report in China<sup>14</sup>.

The current study aims to investigate the distribution percentage of *Cryptosporidium* infection in camels. Due to their physiological attributes, camels are the most suitable species of domestic mammals for thriving in extremely arid conditions<sup>15,16</sup>. The objective of this research is to determine the prevalence of *Cryptosporidium* infection in camels, considering the large population of camels in the western areas of Iraq and their physiological adaptations for survival in arid conditions. The study involves the isolation and diagnosis of *Cryptosporidium parvum* using a real-time PCR technique, taking into account the sex and age of the camels.

**A**pproximately 10 grams of feces was collected from the rectum for each sample in a sterilized cup. The study was conducted on 50 camels living in Al-Qadisiyah province, Iraq, during the period from November 2022 to April 2023, including 25 samples for each sex. According to age, the number of young camels (age less than one year) was 14, between 1–2 years, 19 samples as young, and 17 samples as adults aged more than 2 to 10 years old.

## Molecular diagnosis

### Genomic DNA Extraction

DNA extraction was performed using a stool DNA extraction Kit from Bioneer company, made in Korea, following the company's instructions. The extracted DNA was then tested using a Nanodrop spectrophotometer and stored at -20°C.

### Real-Time PCR technique

Real-Time PCR was conducted using TaqMan qPCR amplification to detect the heat shock protein gene in *C. parvum*. The probes and primers were designed in the research using the NCBI-GenBank recorded sequence for *C. parvum* isolate 11 heat shock protein 70 (HSP70) gene (GenBank: AF221528.1) and designed online using primer3 plus. The primers were provided by Bioneer, made in Korea, as shown in Table 1."

**Table 1. heat shock protein gene (hsp) primers and probe.**

| Primer     | Sequence (5'-3')                   |                      | Amplicon |
|------------|------------------------------------|----------------------|----------|
| hsp primer | F                                  | AATGACTCACAGCGTCAAGC | 96bp     |
|            | R                                  | AGCAATAGCAGCTGCAGTTG |          |
| hsp probe  | FAM- TGCAGGTGCAATTGCTGGTTTGA -BHQ1 |                      |          |

Amplification reaction of RT-PCR was done by kit called (AccuPower® DualStar™ qPCR PreMix kit, Bioneer Company. Made in Korea) and the master mix were compound depending on the manufacturer directions as (table 2).

**Table 2. Real-Time PCR master mix preparation.**

| RT-PCR master mix    | Volume         |
|----------------------|----------------|
| Genomic DNA template | (2) $\mu$ L    |
| F primer (10pmol)    | (1) $\mu$ L    |
| R primer (10pmol)    | (1) $\mu$ L    |
| Hsp probe (20pmol)   | (2.5) $\mu$ L  |
| Free nuclease water  | (13.5) $\mu$ L |
| Total volume         | (20) $\mu$ L   |

Components of the master mix reaction listed in the table (2), it was kept in qPCR tubes then shake by vortex for three minutes, after that, place in RT-PCR system as (table 3).

**Table 3. Real-Time PCR thermocycler conditions**

| RT-PCR step          | Temperature | Time     | Repeat cycle |
|----------------------|-------------|----------|--------------|
| Initial Denaturation | 95 °C       | 5 minute | 1            |
| Denaturation         | 95 °C       | 30 sec   | 45           |
| Annealing\ Extension | 58 °C       | 1min     |              |
| Detection(scan)      |             |          |              |

The data is analyzed by using SPSS software, (Q-test) was applied for compression between the variables at ( $P$ -value <0.05).

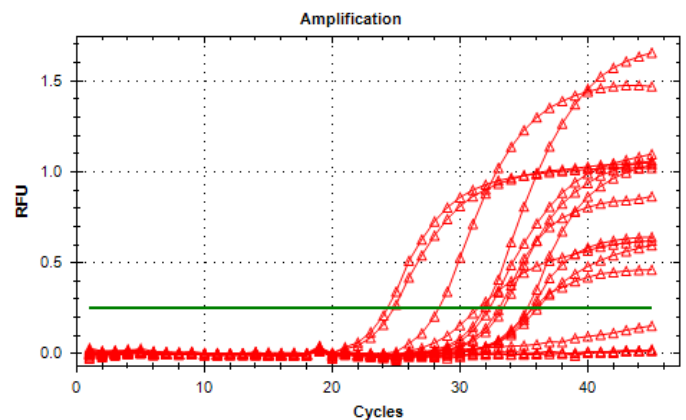
### Statistical Analysis

The software SPSS was used for statistical analysis. The association between infection rate of *Cryptosporidium* and each of sex and age was detected using Chisquare test, Probability values of  $P$  <0.05 were considered statistically significant.

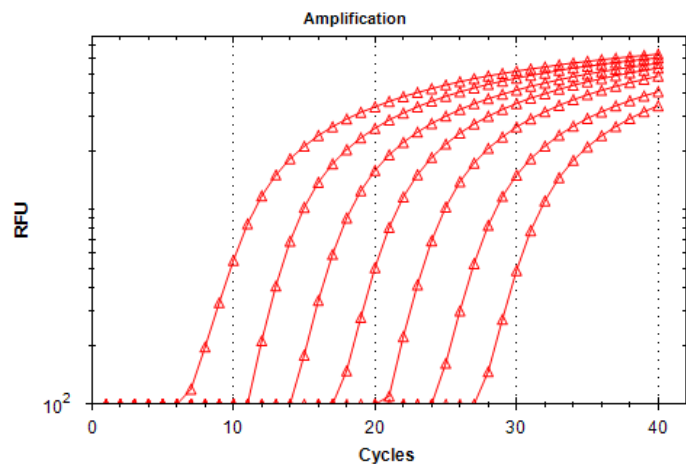
## Results

The study results have demonstrated that, out of a total of 50 cardiac tissue samples, 12 were found to test positively for specific cardiac biomarkers using the direct Real-Time PCR technique, indicating an infection rate of 24%. Real-Time PCR analysis revealed that the positive samples exhibited threshold cycle (CT value) values ranging between 24.29 and 35.69 (see Figure 1). To assess the efficiency of Real-Time PCR amplification with the TaqMan probe (FAM), a standard curve was constructed using these positive samples. The results indicate that the qPCR efficiency can reach up to 99.9%, as illustrated in Figures 2 and 3.

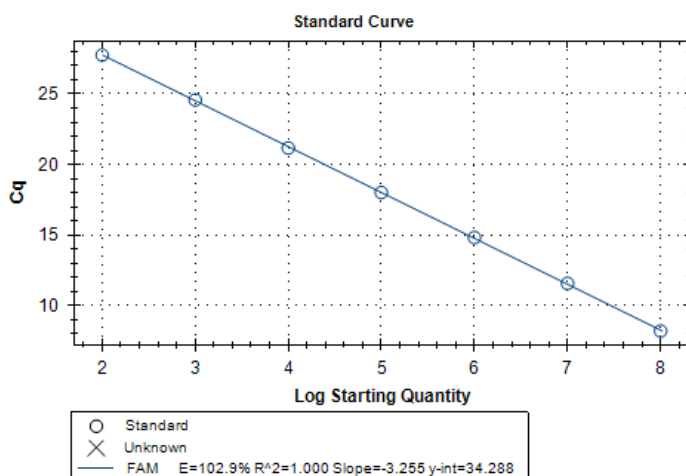
**Figure 1. Real-Time PCR amplification plots that show the positive amplification of the stx1 gene in Shiga toxin-producing Escherichia coli (STEC) isolates. The isolates which appeared amplification at 15 cycles contained a very large amount of DNA Escherichia coli (STEC). The amplification have appeared later, samples were negative which appeared under the threshold line.**



**Figure 2. Real-Time PCR amplification plots that show the amplification of positive control STEC isolate for standard curve of stx1 gene that show specificity of Real-Time PCR based TaqMan probe (FAM) in four serial DNA dilution.**



**Figure 3. Real-Time PCR standard curve that shows the log amplification stx1 gene in positive control four serial DNA dilution of STEC isolate that show qPCR efficiency (99.9%).**



Investigating the pathophysiology of the heart in relation to age, significant differences were noted ( $p=0.048$ ). The results revealed that out of 14 samples from individuals less than one year old, 2 showed positive findings. Similarly, among the 19 samples from the age group of 1-2 years, 2 tested positive for heart-related conditions. However, among the adult population, 8 out of 9 samples showed positive results, indicating a notably higher infection rate at the adult stage, reaching 47.05%. In comparison, the less than one year age group exhibited a 14.28% infection rate, and the young stage had a 10.52% infection rate, respectively. These findings are summarized in Table 4.

**Table 4. Prevalence rate of *C. parvum* according to age**

| qPCR * Age Crosstabulation |          |                 |       |             |       |               |       |       |
|----------------------------|----------|-----------------|-------|-------------|-------|---------------|-------|-------|
|                            |          | Age             |       |             |       |               |       |       |
|                            |          | < than one year |       | Young 1-2 Y |       | Adult >2-10 Y |       | Total |
| qPCR                       | negative | 12              | 85.71 | 17          | 89.47 | 9             | 52.94 | 38    |
|                            | positive | 2               | 14.28 | 2           | 10.52 | 8             | 47.05 | 12    |
| Total                      |          | 14              |       | 19          |       | 17            |       | 50    |

Chi-square: 2.0

$P=0.048$

\* $P<0.05$  Significant

In table 5 significant differences  $p=0.043$  association with gender, the results indicate that 4 positive samples against 20 negative samples for males and 8 positive samples against 18 negative samples for females were recorded. The highly infection rate 30.76% was observed in female while in male it was 16.66%.

**Table 5. Infection rate of *C. parvum* according to gender**

| qPCR * Sex Crosstabulation |          |      |       |        |       |       |     |
|----------------------------|----------|------|-------|--------|-------|-------|-----|
|                            |          | sex  |       |        |       |       |     |
|                            |          | Male | %     | Female | %     | Total | %   |
| qPCR                       | negative | 20   | 83.33 | 18     | 69.23 | 38    | 76  |
|                            | positive | 4    | 16.66 | 8      | 30.76 | 12    | 24  |
| Total                      |          | 24   |       | 26     |       | 50    | 100 |

\* $P<0.05$  Significant

Chi-square: 2.134

$P$  value: 0.043

## Discussion

One of the critical challenges faced by breeders in camel husbandry is the disease caused by *Cryptosporidium* protozoan parasites, which can have a significant impact on the physiology of the camel's heart. This disease not only leads to a reduction in milk and meat production but also impairs the overall performance of camels, potentially resulting in fatal consequences. Moreover, there is a zoonotic concern associated with the transmission of this disease to humans.

According to the findings of a recent study, the prevalence rate of *Cryptosporidium* infection in camels was 24%, which is relatively consistent with a previous study conducted by Sazmand et al. in Iran, where they reported an infection rate of 20.33%. Similarly, studies in Egypt reported infection rates ranging between 17.5% and 19.3%. However, there was a significant discrepancy with a study carried out in Najef, Iraq, in 2015, which reported an exceptionally high infection rate of 61%. Another study by Jawad and Jasim in 2016 also reported a high infection rate of 60%. These differences in infection rates across various regions may be attributed to variations in environmental conditions and hygiene practices. Additionally, variations in diagnostic methods play a crucial role, as the use of optical microscopes, which have lower sensitivity and specificity, can lead to differing results compared to molecular assay-based diagnostic methods.

Regarding the age-related distribution of *C. parvum* infection, the data indicate non-significant differences among different age groups. The highest infection rate of 47.05% was observed in adult camels, which were asymptomatic carriers of the disease and a potential source of infection for young camels. This can be attributed to the more developed immune system in adult camels. Furthermore, the feeding habits and grazing in open areas expose adult camels to other infected animals, which can serve as sources of infection. This pattern is consistent with a study in Iran that reported



a higher prevalence rate of *Cryptosporidium* species in adult camels. In Najef province, Iraq, adult camels between 3 and 6 years old recorded a 75% infection rate. However, there was a discrepancy with the results of a study that found a 20% infection rate in camels less than one year old, compared to 6.5% in adults.

Regarding the sex-related distribution of *Cryptosporidium parvum* infection, there were significant differences between males and females. Female camels showed a higher infection rate of 30.76% compared to males, who had an infection rate of 16.66%. This difference may be attributed to the traditional gender roles in animal husbandry, where females are typically not used for labor-intensive tasks, unlike males. These findings are not consistent with a study that reported infection rates of 63.63% for males and 59.70% for females. Another study also reported higher infection rates for males (61.16%) compared to females (38.83%). However, some studies have concluded that there is no significant difference in the prevalence of *Cryptosporidium*-like infections between male and female camels.

In conclusion, a profound grasp of the pathophysiology of the heart is of utmost importance, as the heart plays a pivotal role in sustaining human health and well-being. The heart's intricate mechanisms are not only responsible for pumping oxygenated blood throughout the body but also for maintaining the delicate balance of various physiological processes. Any disruptions in the normal functioning of the heart can lead to a cascade of health issues and, in severe cases, prove fatal. The heart is a marvel of biological engineering, with a complex network of cardiac muscles, electrical impulses, and valves that work in harmony to ensure a continuous flow of blood to every cell and tissue in the body. It is responsible for oxygen and nutrient delivery and waste product removal. Any deviations from its precise operation can have profound consequences, impacting not only an individual's quality of life but also their longevity. Cardiovascular diseases are a leading cause of morbidity and mortality worldwide, making the understanding of heart pathophysiology even more critical. These conditions include a wide range of disorders, such as coronary artery disease, congestive heart failure, arrhythmias, and congenital heart defects. Effective prevention, early diagnosis, and appropriate management of these conditions are paramount in reducing the global burden of cardiovascular disease. Thus, it is imperative to establish comprehensive programs aimed at both preventing heart-related disorders and managing them effectively. These programs should encompass public health ini-

tiatives, medical research, and healthcare policies designed to educate individuals about heart health, promote healthy lifestyles, and ensure access to state-of-the-art diagnostic and treatment modalities. By addressing heart pathophysiology with diligence and dedication, we can enhance the quality of life for individuals, reduce healthcare costs, and ultimately save countless lives.

### Acknowledgments

Not applied.

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